SIEVED MIRROR KERNELS, GONEK–HUGHES–KEATING VERSUS ENTIRE EXPONENTIAL PRIME FACTORS, PRIME-POWER PURITY ON $(X, X^2]$, AND FUNCTIONAL-EQUATION OBSTRUCTIONS

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Abstract. We study finite Dirichlet multipliers $M_X(s)$ acting on Riemann's Ξ-function under canonical normalization $W(d,t) = \frac{1}{2}\partial_d|\Xi(t-\mathrm{i}d)|^2 = 16\int_0^\infty y\sinh(2dy)I(y,t)\,\mathrm{d}y$, showing that every finite real multiplier generates an even sieved kernel $\Phi_X(u)$ holomorphic on $|\mathrm{Im}(u)| < \frac{\pi}{8}$. On $\sigma > 1$, comparing Gonek–Hughes–Keating's factor $P_X(s)^{-1} = \prod_{p\le X}(1-p^{-s})$ against the entire factor $E_X(s) = \exp(-\sum_{2\le n\le X}\frac{\Lambda(n)}{\ln n}n^{-s})$ proves that $\zeta(s)E_X(s) = \sum_{N=1}^\infty b_N^{\mathcal{E}}(X)N^{-s}$ satisfies $b_1^{\mathcal{E}} = 1$, $b_N^{\mathcal{E}} = 0$ on $[2,\lfloor X\rfloor]$, $0 \le b_N^{\mathcal{E}} \le 1$, and prime-power purity $b_N^{\mathcal{E}}(X) = \Lambda(N)/\ln N$ on $(\lfloor X\rfloor, (\lfloor X\rfloor+1)^2)$. In $0 < \sigma \le 1$, we prove three obstructions: $\mathcal{P}_{>X}(s)$ diverges and $E_X(s)$ lacks a functional equation so $\zeta(s)M_X(s)$ leaves the reflected Riemann–Siegel branch unsieved; setting $X = N(t)$ makes $\theta < \frac{1}{2}$ tautological while X^{-2d} stalls at $d \asymp 1/\ln t$; and single-line weights $w_n^*(d_0) = -(n^{2d_0}-1)^{-1}$ diverge as $d_0 \to 0^+$ against Selberg's explicit-formula zero sum.

1. Introduction and Main Results

In Riemann's integral representation [9, 5, 8, 13], $\Xi(z) := \xi(\frac{1}{2}+\mathrm{i}z) = 8\int_0^\infty \Phi(u)\cos(2zu)\,\mathrm{d}u$ ($z = t-\mathrm{i}d$, $s = \frac{1}{2}+d+\mathrm{i}t$) has canonical transverse Wronskian [12, 6, 3, 1, 14, 4]

$$W(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\big|\Xi(t-\mathrm{i}d)\big|^2 = R(d,t)I_t(d,t) - I(d,t)R_t(d,t) = 16\int_0^\infty y\sinh(2dy)\,I(y,t)\,\mathrm{d}y, \tag{1.1}$$

with $I(y,t) := \int_{-\infty}^\infty \Phi(\frac{x+y}{2})\Phi(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x$. As shown in Paper V [7], while individual shells $\phi_m(u)$ of $\Phi(u) = \sum_{m=1}^\infty \phi_m(u)$ exhibit Archimedean branch damping, the $\binom{N(t)}{2}$ off-diagonal Dirichlet cross-pairs sum in L^1 to $\asymp t^{1/2}$, reflecting the fact that $\Phi(u)$ does not use the Euler product $\zeta(s) = \prod_p(1-p^{-s})^{-1}$.

A classical strategy in analytic number theory is to dampen low-integer oscillations by multiplying $\zeta(s)$ by a finite prime-supported Dirichlet factor. In their hybrid Euler–Hadamard product, Gonek, Hughes, and Keating [2] studied the truncated Euler product reciprocal

$$P_X(s)^{-1} := \prod_{p\le X}(1-p^{-s}) = \sum_{P^+(k)\le X}\mu(k)\,k^{-s}, \tag{1.2}$$

whereas Selberg's mollifier method [10] damps $\zeta(\frac{1}{2}+\mathrm{i}t)$ in mean square via short Dirichlet polynomials of length $X \le T^{1/2-\varepsilon}$. Alternatively, truncating the Dirichlet series of $-\ln\zeta(s)$

Date: November 2026.

2020 *Mathematics Subject Classification.* Primary 11M06, 11N36; Secondary 30D15, 41A60.

Key words and phrases. Riemann Ξ-function, transverse Wronskian, Gonek–Hughes–Keating hybrid product, sieved Jacobi kernel, prime-power purity, Selberg explicit formula.

at integer horizon $n \le X$ defines the entire zero-free exponential factor

$$E_X(s) := \exp\left(- \sum_{2 \le n \le X} \frac{\Lambda(n)}{\ln n}\, n^{-s} \right) = P_X(s)^{-1} \exp\left(\sum_{p \le X} \sum_{p^m > X} \frac{p^{-ms}}{m} \right). \tag{1.3}$$

Analyzing how $P_X(s)^{-1}$, $E_X(s)$, and their reflection-symmetrized counterparts act on $\Xi(z)$ requires strictly separating genuine identities on $\sigma > 1$ from invalid formal manipulations inside the critical strip $0 < \sigma \le 1$.

Theorem 1.1 (Sieved Kernel Identities and Critical-Strip Obstructions)**.** *Let $X \ge 2$, $z = t - \mathrm{id}$, and $s = \frac{1}{2} + d + \mathrm{i}t$.*

(i) ***Strip-Holomorphic Sieved Mirror Kernel.*** *For any finite real Dirichlet polynomial $M_X(s) = \sum_{k \in \mathcal{I}_X} c_k k^{-s}$, the symmetric multiplier $\mathcal{M}_X(z) := M_X(\frac{1}{2} + \mathrm{i}z) M_X(\frac{1}{2} - \mathrm{i}z)$ yields $\Xi_X(z) := \Xi(z)\mathcal{M}_X(z) = 8 \int_0^\infty \Phi_X(u) \cos(2zu)\, \mathrm{d}u$, where $\Phi_X(u)$ in (2.1) is even and holomorphic throughout $|\mathrm{Im}(u)| < \frac{\pi}{8}$.*

(ii) ***Gonek–Hughes–Keating vs. Entire Sieve Purity on $\sigma > 1$.*** *For $\mathrm{Re}(s) > 1$, $\zeta(s)E_X(s) = \sum_{N=1}^\infty b_N^{\mathcal{E}}(X) N^{-s}$ converges absolutely with $b_1^{\mathcal{E}}(X) = 1$, $b_N^{\mathcal{E}}(X) = 0$ for $2 \le N \le \lfloor X \rfloor$, $0 \le b_N^{\mathcal{E}}(X) \le 1$ for all $N \ge 1$, and exact prime-power purity $b_N^{\mathcal{E}}(X) = \Lambda(N)/\ln N$ on $\lfloor X \rfloor < N < (\lfloor X \rfloor + 1)^2$, differing from $b_N^{\mathrm{GHK}}(X) = \mathbf{1}_{P^-(N) > X}$ precisely at prime powers $p^k > X$ of primes $p \le X$. Moreover, truncating $E_X(s)$ at $k \le M$ preserves every coefficient $b_N^{\mathcal{E}}(X)$ for $1 \le N \le M$.*

(iii) ***Critical-Strip Obstructions ($0 < \sigma \le 1$).*** *Inside $0 < \sigma \le 1$: (a) $\mathcal{P}_{>X}(s) = \sum_{n > X} \frac{\Lambda(n)}{\ln n} n^{-s}$ diverges and $E_X(s)$ is zero-free without a functional equation, so $\zeta(s)E_X(s)$ has the exact same zeros as $\zeta(s)$ and $\zeta(s)M_X(s)$ leaves both a convolution tail on $(N(t), N(t)K]$ and an unsieved reflected branch $\chi(s)M_X(s)\sum_{m \le N(t)} m^{s-1}$; (b) setting $X = N(t)$ in a formal $N(t)$-term sum makes $\theta = \ln X / \ln t < \frac{1}{2}$ tautological, while $\frac{X^{-2d}}{2d \ln X} \asymp 1$ fails to decay at zero-detection depth $d \asymp 1/\ln t$; and (c) single-line weights $w_n^*(d_0) = -(n^{2d_0} - 1)^{-1}$ fail for $d \ne d_0$, diverge as $-(2d_0 \ln n)^{-1} \to -\infty$ as $d_0 \to 0^+$, and cannot cancel the nontrivial zero poles in Selberg's explicit formula [11].*

1.1. **Proof overview.** The proofs proceed in three steps:

(1) *Real shift invariance of $|\mathrm{Im}(u)| < \frac{\pi}{8}$ (Section 2).* Expanding $\cos(2zu)\cos(z\ln(k/m))$ translates $\Phi(u)$ by real shifts $\pm\frac{1}{2}\ln(k/m) \in \mathbb{R}$, preserving $\mathrm{Im}(u)$ and the canonical Wigner representation.

(2) *Local Euler generating functions on $\sigma > 1$ (Section 3).* Factoring $\zeta(s)E_X(s)$ prime-by-prime into $(1 - p^{-s})^{-1} \exp(-\sum_{m=1}^{k_p} p^{-ms}/m)$ proves universal $[0,1]$ contractivity via $(1 - z)^{-1}\exp(-\sum_{m=1}^k z^m/m)$ and isolates single prime powers below $(\lfloor X \rfloor + 1)^2$.

(3) *Approximate functional equation and explicit-formula decomposition in $0 < \sigma \le 1$ (Sections 4–5).* Multiplying the Riemann–Siegel decomposition of $\zeta(s)$ by $M_X(s)$ exhibits the mixed exponents $k^{-s}m^{s-1}$ on the reflected branch, while Selberg's explicit formula [11] exhibits the residue poles $(\rho - s)^{-1}$ missed by formal Dirichlet differentiation.

1.2. **Notation and canonical normalizations.** Throughout, $s = \sigma + \mathrm{i}t = \frac{1}{2} + d + \mathrm{i}t$, $z = t - \mathrm{id}$, $\Xi(z) = \xi(\frac{1}{2} + \mathrm{i}z)$, $N(t) = \lfloor \sqrt{t/2\pi} \rfloor$, and $\chi(s) = \pi^{s-1/2}\Gamma(\frac{1-s}{2})/\Gamma(\frac{s}{2})$. For $N \ge 2$, $P^-(N)$ and $P^+(N)$ denote the smallest and largest prime factors of N ($P^-(1) := +\infty$, $P^+(1) := 1$), $\omega(N)$ denotes the number of distinct prime factors of N, and $\Lambda(n)$ denotes the von Mangoldt function.

2. Strip-Holomorphic Sieved Mirror Kernels

Lemma 2.1 (Strip Preservation Under Finite Real Translations)**.** *Let $\Phi(u)$ be even on $\mathbb{R}$ and holomorphic in the open strip $\mathcal{S}_{\pi/8} := \{u \in \mathbb{C} : |\mathrm{Im}(u)| < \frac{\pi}{8}\}$. For any finite index set $\mathcal{I}_X \subset \mathbb{N}$ and real weights $\{c_k\}_{k\in\mathcal{I}_X}$, the function*

$$\Phi_X(u) := \frac{1}{2}\sum_{k,m\in\mathcal{I}_X} \frac{c_k c_m}{\sqrt{km}}\left[\Phi\Big(u+\frac{1}{2}\ln\frac{k}{m}\Big)+\Phi\Big(u-\frac{1}{2}\ln\frac{k}{m}\Big)\right] \tag{2.1}$$

is even ($\Phi_X(-u)=\Phi_X(u)$), holomorphic throughout $\mathcal{S}_{\pi/8}$, and maintains super-exponential decay $\mathcal{O}(\exp(-\gamma \mathrm{e}^{4|\mathrm{Re}(u)|}))$ as $|\mathrm{Re}(u)|\to\infty$ on every closed substrip $|\mathrm{Im}(u)|\le\eta<\frac{\pi}{8}$.

Proof. Since $\delta_{k,m} := \frac{1}{2}\ln(k/m) \in \mathbb{R}$, $\mathrm{Im}(u\pm\delta_{k,m}) = \mathrm{Im}(u)$, so $u\in\mathcal{S}_{\pi/8} \iff u\pm\delta_{k,m}\in\mathcal{S}_{\pi/8}$. Because $\mathcal{I}_X$ is finite, the finite linear combination (2.1) is holomorphic on $\mathcal{S}_{\pi/8}$ and inherits super-exponential decay with $\gamma = \pi\min_{k,m}(m/k)\cos(4\eta) > 0$. Replacing u by $-u$ swaps $u+\delta_{k,m}$ with $-(u-\delta_{k,m})$, which leaves (2.1) invariant by $\Phi(-w)=\Phi(w)$. □

Proposition 2.2 (Canonical Sieved Wronskian Representation)**.** *Let $M_X(s) := \sum_{k\in\mathcal{I}_X} c_k k^{-s}$ ($c_k\in\mathbb{R}$, $|\mathcal{I}_X|<\infty$) and $\mathcal{M}_X(z) := M_X(\frac{1}{2}+\mathrm{i}z)M_X(\frac{1}{2}-\mathrm{i}z)$. Then $\Xi_X(z) := \Xi(z)\mathcal{M}_X(z) = 8\int_0^\infty \Phi_X(u)\cos(2zu)\,\mathrm{d}u$, and its canonical transverse Wronskian satisfies*

$$\boxed{W_X(d,t) := \frac{1}{2}\frac{\partial}{\partial d}\big|\Xi_X(t-\mathrm{i}d)\big|^2 = R_X I_{X,t} - I_X R_{X,t} = 16\int_0^\infty y\sinh(2dy)\,I_X(y,t)\,\mathrm{d}y,} \tag{2.2}$$

where $I_X(y,t) := \int_{-\infty}^\infty \Phi_X(\frac{x+y}{2})\Phi_X(\frac{x-y}{2})\cos(2tx)\,\mathrm{d}x$.

Proof. Writing $\mathcal{M}_X(z) = \sum_{k,m\in\mathcal{I}_X}\frac{c_kc_m}{\sqrt{km}}\mathrm{e}^{-\mathrm{i}z\ln(k/m)}$ and pairing (k,m) with (m,k) yields $\mathcal{M}_X(z) = \sum_{k,m\in\mathcal{I}_X}\frac{c_kc_m}{\sqrt{km}}\cos(z\ln\frac{k}{m})$. Multiplying $\Xi(z) = 4\int_{-\infty}^\infty\Phi(v)\cos(2zv)\,\mathrm{d}v$ by $\cos(2z\delta_{k,m})$, using $2\cos(2zv)\cos(2z\delta) = \cos(2z(v-\delta))+\cos(2z(v+\delta))$, and setting $u = v\mp\delta_{k,m}$ gives $\Xi_X(z) = 4\int_{-\infty}^\infty\Phi_X(u)\cos(2zu)\,\mathrm{d}u = 8\int_0^\infty\Phi_X(u)\cos(2zu)\,\mathrm{d}u$. Applying the Cauchy–Riemann identity and Wigner coordinates $(x,y)=(u+v,u-v)$ as in (1.1) yields (2.2). □

Remark 2.3 (Zeros introduced by $P_X(s)^{-1}$ versus zero-freeness of $E_X(s)$). For Gonek–Hughes–Keating's factor $M_X(s)=P_X(s)^{-1}=\prod_{p\le X}(1-p^{-s})$, $\mathcal{I}_X$ has $|\mathcal{I}_X|=2^{\pi(X)}$ squarefree terms, so Lemma 2.1 applies directly; however, $P_X(s)^{-1}$ vanishes on $\sigma=0$ at $s=\frac{2\pi\mathrm{i}j}{\ln p}$ ($p\le X, j\in\mathbb{Z}$), so $\mathcal{M}_X(z)$ introduces lines of extraneous zeros at $d=\pm\frac{1}{2}$. By contrast, $E_X(s)$ in (1.3) is entire and zero-free on $\mathbb{C}$, and its finite Dirichlet truncations $M^{\mathcal{E}}_{X,M}(s)=\sum_{k=1}^M c_k^{\mathcal{E}}(X)k^{-s}$ satisfy Lemma 2.1 for every $M<\infty$ (whereas the infinite series of $E_X(s)$ itself cannot be interchanged with Φ without bounding arbitrarily large shifts $\frac{1}{2}\ln(k/m)$).

3. Gonek–Hughes–Keating versus Entire Prime Factors on $\sigma>1$

On the half-plane of absolute convergence $\sigma=\mathrm{Re}(s)>1$, $-\ln\zeta(s) = -\sum_{n=2}^\infty\frac{\Lambda(n)}{\ln n}n^{-s}$ converges absolutely, allowing an exact arithmetic comparison between $\zeta(s)P_X(s)^{-1}$ [2] and $\zeta(s)E_X(s)$.

Theorem 3.1 (Arithmetic Structure of $\zeta(s)E_X(s)$ versus $\zeta(s)P_X(s)^{-1}$ on $\sigma>1$)**.** *Let $X\ge 2$ and $\mathrm{Re}(s)>1$. Then $\zeta(s)P_X(s)^{-1} = \sum_{N=1}^\infty b_N^{\mathrm{GHK}}(X)N^{-s}$ with $b_N^{\mathrm{GHK}}(X) = \mathbf{1}_{P^-(N)>X}\in\{0,1\}$, whereas $\zeta(s)E_X(s) = \exp(\sum_{n>X}\frac{\Lambda(n)}{\ln n}n^{-s}) = \sum_{N=1}^\infty b_N^{\mathcal{E}}(X)N^{-s}$ converges absolutely and satisfies:*

(i) ***Exact Low-Integer Annihilation.*** *$b_1^{\mathcal{E}}(X)=1$ and $b_N^{\mathcal{E}}(X)=0$ for all $2\le N\le\lfloor X\rfloor$.*

TABLE 1. 50-decimal verification of Theorem 3.1 at sieve horizon $X = 10$ (quadratic purity window $10 < N < 121$): comparison of $b_N^{\mathcal{E}}(10)$, Gonek–Hughes–Keating $b_N^{\mathrm{GHK}}(10)$, and finite-divisor sum $\sum_{k|N} c_k^{\mathcal{E}}(10)$.

Index N	Arithmetic Type	$b_N^{\mathcal{E}}(10)$	$b_N^{\mathrm{GHK}}(10)$	$\lvert b_N^{\mathcal{E}} - \sum_{k\mid N} c_k^{\mathcal{E}}\rvert$	Regime
1	Unit	1.0000000000	1	0.00×10^{0}	Exact unit
2–10	Sieved $N \le X$	0.0000000000	0	0.00×10^{0}	Annihilated $[2, X]$
$11, 13, 17$	Primes $p > X$	1.0000000000	1	0.00×10^{0}	Pure primes in $(X, 121)$
$16 = 2^4$	Prime power $p \le X$	0.2500000000	0	2.41×10^{-52}	Fractional $1/k$ price
$27 = 3^3$	Prime power $p \le X$	0.3333333333	0	1.93×10^{-52}	Fractional $1/k$ price
$77 = 7 \cdot 11$	Composite in $(X, 121)$	0.0000000000	0	1.54×10^{-52}	$\omega(N) \ge 2$ zero
$121 = 11^2$	First $r \ge 2$ onset	1.0000000000	1	0.00×10^{0}	$P^-(121) = 11 > 10$
$143 = 11 \cdot 13$	Rough semiprime	1.0000000000	1	0.00×10^{0}	$P^-(143) > 10$

(ii) ***Quadratic Prime-Power Purity Window.*** *For every integer* $\lfloor X \rfloor < N < (\lfloor X \rfloor + 1)^2$,

$$b_N^{\mathcal{E}}(X) = \frac{\Lambda(N)}{\ln N} = \begin{cases} 1/k, & \text{if } N = p^k \text{ is a prime power } (k \ge 1), \\ 0, & \text{if } \omega(N) \ge 2. \end{cases} \tag{3.1}$$

In particular, on $(\lfloor X \rfloor, (\lfloor X \rfloor + 1)^2)$, $b_N^{\mathcal{E}}(X) - b_N^{\mathrm{GHK}}(X) = \frac{1}{k}$ *if* $N = p^k > X$ *with* $p \le X$ $(k \ge 2)$, *and* 0 *otherwise.*

(iii) ***Local Multiplicative Generating Function and Universal*** $[0, 1]$ ***Contractivity.*** *For every* $N = \prod_i p_i^{a_i} \ge 1$, $b_N^{\mathcal{E}}(X) = \prod_i \beta_{a_i}(k_{p_i})$ *where* $k_p := \lfloor \frac{\ln X}{\ln p} \rfloor$ $(k_p = 0$ *for* $p > X)$ *and* $\sum_{a=0}^{\infty} \beta_a(k) z^a := (1 - z)^{-1} \exp(-\sum_{m=1}^{k} \frac{z^m}{m})$. *Every local coefficient satisfies* $0 \le \beta_a(k) \le 1$, *so* $0 \le b_N^{\mathcal{E}}(X) \le 1$ *for all* $N \ge 1$, *with* $b_N^{\mathcal{E}}(X) = 1$ *whenever* $P^-(N) > X$.

(iv) ***Exact Finite-Divisor Truncation.*** *Writing* $E_X(s) = \sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X) k^{-s}$ *and* $M_{X,M}^{\mathcal{E}}(s) := \sum_{k=1}^{M} c_k^{\mathcal{E}}(X) k^{-s}$, *the Dirichlet product* $\zeta(s) M_{X,M}^{\mathcal{E}}(s) = \sum_{N=1}^{\infty} b_{N,M}^{\mathcal{E}}(X) N^{-s}$ *satisfies* $b_{N,M}^{\mathcal{E}}(X) = b_N^{\mathcal{E}}(X)$ *identically for all* $1 \le N \le M$.

Proof. For $\mathrm{Re}(s) > 1$, $\zeta(s) P_X(s)^{-1} = \prod_{p > X} (1 - p^{-s})^{-1} = \sum_{P^-(N) > X} N^{-s}$, while $\ln \zeta(s) = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{\ln n} n^{-s}$ gives $\zeta(s) E_X(s) = \exp(\mathcal{P}_{>X}(s))$ with $\mathcal{P}_{>X}(s) := \sum_{n > \lfloor X \rfloor} \frac{\Lambda(n)}{\ln n} n^{-s}$.

(i)–(ii) Expanding $\exp(\mathcal{P}_{>X}(s)) = 1 + \mathcal{P}_{>X}(s) + \sum_{r=2}^{\infty} \Gamma(r+1)^{-1} \mathcal{P}_{>X}(s)^r$, every term in $\mathcal{P}_{>X}(s)$ has $n \ge \lfloor X \rfloor + 1$, so $b_1^{\mathcal{E}}(X) = 1$ and $b_N^{\mathcal{E}}(X) = 0$ for $2 \le N \le \lfloor X \rfloor$. Every term in $\mathcal{P}_{>X}(s)^r$ for $r \ge 2$ is a product $n_1 \cdots n_r \ge (\lfloor X \rfloor + 1)^2$. Hence on $\lfloor X \rfloor < N < (\lfloor X \rfloor + 1)^2$, only the linear term $r = 1$ contributes, giving $b_N^{\mathcal{E}}(X) = \Lambda(N) / \ln N$. Since $b_N^{\mathrm{GHK}}(X) = 1$ on primes $p > X$ and 0 on all composites and prime powers p^k of $p \le X$ inside $(X, (\lfloor X \rfloor + 1)^2)$, their difference is supported purely on prime powers $p^k > X$ with $p \le X$.

(iii) Factoring $\zeta(s) E_X(s) = \prod_p [(1 - p^{-s})^{-1} \exp(-\sum_{m=1}^{k_p} \frac{p^{-ms}}{m})]$ yields $b_N^{\mathcal{E}}(X) = \prod_i \beta_{a_i}(k_{p_i})$. On $|z| < 1$, $(1 - z)^{-1} \exp(-\sum_{m=1}^{k} \frac{z^m}{m}) = \exp(\sum_{m=k+1}^{\infty} \frac{z^m}{m})$ has non-negative power-series coefficients, so $\beta_a(k) \ge 0$. Comparing coefficientwise with $\exp(\sum_{m=1}^{\infty} \frac{z^m}{m}) = (1 - z)^{-1} = \sum_{a=0}^{\infty} z^a$ gives $\beta_a(k) \le 1$ for all $a \ge 0$, with $\beta_a(0) = 1$.

(iv) In $\zeta(s) E_X(s) = (\sum_{m=1}^{\infty} m^{-s})(\sum_{k=1}^{\infty} c_k^{\mathcal{E}}(X) k^{-s})$, the N-th coefficient is $\sum_{k|N} c_k^{\mathcal{E}}(X)$. For $1 \le N \le M$, every divisor $k \mid N$ satisfies $k \le N \le M$, so $b_{N,M}^{\mathcal{E}}(X) = \sum_{k|N, k \le M} c_k^{\mathcal{E}}(X) = b_N^{\mathcal{E}}(X)$. □

4. Critical-Strip Obstructions in $0 < \sigma \le 1$

We now prove why the $\sigma > 1$ identities of Theorem 3.1 cannot be imported naively into the critical strip $0 < \sigma \le 1$ to deduce zero-freeness or sub-Selberg mollification.

Proposition 4.1 (Failure of $\exp(\mathcal{P}_{>X}(s))$ and Lack of a Functional Equation)**.** *Let $0 < \sigma = \frac{1}{2} + d \le 1$ and $X \ge 2$.*

(i) ***Divergence of $\mathcal{P}_{>X}(s)$ and Zero Invariance.*** *On $0 < \sigma \le 1$, the Dirichlet series $\mathcal{P}_{>X}(s) = \sum_{n>X} \frac{\Lambda(n)}{\ln n} n^{-s}$ diverges unconditionally (its conditional convergence for $\sigma > \frac{1}{2}$ is equivalent to the Riemann hypothesis), and $\ln \zeta(s)$ has logarithmic branch points at every zero ρ of $\zeta(s)$. Because $E_X(s)$ in (1.3) is entire and zero-free on $\mathbb{C}$, the meromorphic continuation of $\zeta(s)E_X(s)$ into $0 < \sigma \le 1$ vanishes at the exact same zeros (with the same multiplicities) as $\zeta(s)$.*

(ii) ***Asymmetric Riemann–Siegel Expansion.*** *Neither $E_X(s)$ nor any nontrivial finite Dirichlet polynomial $M_X(s) = \sum_{k=1}^{K} c_k k^{-s}$ ($K \ge 2$, $c_K \ne 0$) satisfies a Riemann-type functional equation. Multiplying the approximate functional equation $\zeta(s) = \sum_{n=1}^{N(t)} n^{-s} + \chi(s) \sum_{m=1}^{N(t)} m^{s-1} + R_{\mathrm{RS}}(s)$ by $M_X(s)$ gives*

$$\zeta(s)M_X(s) = \sum_{N=1}^{N(t)K} \Bigg(\sum_{\substack{nk=N \\ 1\le n\le N(t),\, 1\le k\le K}} c_k \Bigg) N^{-s} + \chi(s) M_X(s) \sum_{m=1}^{N(t)} m^{s-1} + M_X(s) R_{\mathrm{RS}}(s). \quad (4.1)$$

In (4.1), the forward sum extends to $N(t)K$ with incomplete divisor sums for $N(t) < N \le N(t)K$, while the reflected branch $\chi(s) \sum_{k=1}^{K} \sum_{m=1}^{N(t)} c_k k^{-s} m^{s-1}$ mixes k^{-s} with m^{s-1} rather than sieving m^{s-1} into $\sum b_m m^{s-1}$.

Proof. (i) By Landau's theorem on Dirichlet series with non-negative coefficients, the series $\sum_{n>X} \frac{\Lambda(n)}{\ln n} n^{-\sigma}$ diverges at $\sigma = 1$ (since $\zeta(s)$ has a pole at $s = 1$), and $\ln \zeta(s)$ cannot be continued analytically across any zero ρ of $\zeta(s)$. Since $E_X(s) \ne 0$ on $\mathbb{C}$, $\zeta(s)E_X(s) = 0 \iff \zeta(s) = 0$.

(ii) Multiplying $\zeta(s)$ termwise by $\sum_{k=1}^{K} c_k k^{-s}$ yields (4.1) directly. For $N > N(t)$, divisors $k \mid N$ with $N/k > N(t)$ are absent from the inner sum, breaking Theorem 3.1(iv). On the reflected branch, $k^{-s} m^{s-1} = m^{-1}(m/k)^s$ has phase $+t \ln(m/k)$ and amplitude $k^{-1/2-d} m^{-1/2+d}$, which equals $(mk)^{s-1}$ only for $k = 1$. □

Proposition 4.2 (Tautology of Truncated $\theta < \frac{1}{2}$ and Breakdown at $d \asymp 1/\ln t$)**.** *Let $N(t) = \lfloor \sqrt{t/2\pi} \rfloor$ for $t \ge 14$.*

(i) *In the formal finite sum $S_{\mathrm{formal}}(s; X) := \sum_{n=1}^{N(t)} b_n^{\mathcal{E}}(X) n^{-s}$, choosing $X = N(t)$ gives $b_1^{\mathcal{E}}(N(t)) = 1$ and $b_n^{\mathcal{E}}(N(t)) = 0$ for every $2 \le n \le N(t)$ by Theorem 3.1(i), reducing $S_{\mathrm{formal}}(s; N(t)) \equiv 1$ identically. The logarithmic exponent $\theta(t) := \frac{\ln N(t)}{\ln t} \le \frac{1}{2} - \frac{\ln(2\pi)}{2 \ln t} < \frac{1}{2}$ is therefore an algebraic consequence of $N(t) \le \sqrt{t/2\pi}$, not a bypass of Selberg's mean-square mollifier barrier [10], because S_{formal} omits both the convolution tail on $(N(t), N(t)K]$ and the unsieved reflected branch in (4.1).*

(ii) *For the $\sigma > 1$ integral bound $\int_X^{X^2} \frac{\mathrm{d}u}{u^{1+2d} \ln u} \le \frac{X^{-2d}}{2d \ln X}$ at $X = \sqrt{t/2\pi}$, fixed $d > 0$ gives $O(t^{-d}/\ln t)$ decay, whereas at the critical zero-detection scale $d = c/\ln t$ ($c > 0$) one has $X^{-2d} = \mathrm{e}^{-c}(2\pi)^{c/\ln t} \to \mathrm{e}^{-c}$ and $\frac{X^{-2d}}{2d \ln X} \to \frac{\mathrm{e}^{-c}}{c} = \mathcal{O}(1)$ as $t \to \infty$.*

Proof. Part (i) follows immediately from Theorem 3.1(i) and (4.1). For part (ii), substituting $X = (t/2\pi)^{1/2}$ and $d = c/\ln t$ gives $2d \ln X = c(1 - \frac{\ln(2\pi)}{\ln t}) \to c$ and $X^{-2d} = \exp(-2d \ln X) \to \mathrm{e}^{-c} > 0$. □

Proposition 4.3 (Single-Line Weight Divergence and Selberg's Explicit-Formula Waterbed)**.** *For real weights $\{w_n\}_{2\le n\le X}$, set $\mathcal{E}_{X,w}(z) := \exp(\sum_{2\le n\le X} \frac{w_n\Lambda(n)}{\ln n}[n^{-(1/2+\mathrm{i}z)} + n^{-(1/2-\mathrm{i}z)}])$, so that $-\mathrm{Re}[\mathrm{i}\mathcal{E}'_{X,w}(t-\mathrm{i}d)/\mathcal{E}_{X,w}(t-\mathrm{i}d)] = \sum_{2\le n\le X} w_n\Lambda(n)(n^{-1/2-d}-n^{-1/2+d})\cos(t\ln n)$. Matching the coefficient of $\Lambda(n)\cos(t\ln n)$ to $-n^{-1/2-d_0}$ at a single abscissa $\sigma_0 = \frac12 + d_0 > 1$ uniquely forces $w_n^*(d_0) := -(n^{2d_0}-1)^{-1} < 0$. This choice encounters three strict obstructions:*

(i) ***Single-line restriction:*** *At every $d \neq d_0$, the combined coefficient equals $n^{-1/2-d}[1 - \frac{n^{2d}-1}{n^{2d_0}-1}] \neq 0$.*

(ii) ***Critical-line divergence:*** *As $d_0 \to 0^+$, $w_n^*(d_0) = -\frac{1}{2d_0\ln n} + \frac12 + \mathcal{O}(d_0\ln n) \to -\infty$.*

(iii) ***Explicit-formula zero-sum obstruction on $0 < \sigma \le 1$:*** *Inside the critical strip, $-\zeta'(s)/\zeta(s)$ is not given by $\sum_{n=2}^\infty \Lambda(n)n^{-s}$; by Selberg's smoothed explicit formula [11], for any $\eta \in C_c^\infty([0,\infty))$ satisfying $\eta(u) \equiv 1$ near $u = 0$,*

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=2}^\infty \Lambda(n)\eta\Big(\frac{n}{X}\Big)n^{-s} - \frac{\widetilde{\eta}(1-s)X^{1-s}}{1-s} + \sum_\rho \frac{\widetilde{\eta}(\rho-s)X^{\rho-s}}{\rho-s} + \mathcal{O}_{t,\eta}(X^{-\sigma}) \quad (s \neq 1,\rho), \tag{4.2}$$

where $\widetilde{\eta}(w) := -\int_0^\infty \eta'(u)u^w\,\mathrm{d}u$ is entire with $\widetilde{\eta}(0) = 1$ (so $\int_0^\infty \eta(u)u^{w-1}\,\mathrm{d}u = \widetilde{\eta}(w)/w$ for $\mathrm{Re}(w) > 0$). No entire prime factor $\mathcal{E}_{X,w}(z)$ can cancel the meromorphic poles $(\rho-s)^{-1}$ at nontrivial zeros ρ.

Proof. Solving $w_n(n^{-1/2-d_0} - n^{-1/2+d_0}) = -n^{-1/2-d_0}$ gives $w_n^*(d_0) = -(n^{2d_0}-1)^{-1}$, from which (i) and the Laurent expansion (ii) follow directly. Equation (4.2) follows by integrating $-\frac{\zeta'}{\zeta}(s+w)\frac{\widetilde{\eta}(w)}{w}X^w\frac{\mathrm{d}w}{2\pi\mathrm{i}}$ from $\mathrm{Re}(w) = \max(1, 2-\sigma)$ to $\mathrm{Re}(w) = -\sigma - \frac12$ via Cauchy's residue theorem, picking up the simple poles at $w = 0$, $w = 1-s$, and $w = \rho - s$ [11, 13]. □

5. Shifted Outer Modulus Identity and Zero Invariance

Proposition 5.1 (Exact Shifted Outer Modulus and Wronskian Decomposition)**.** *For $\kappa \ge 0$ and $X \ge 2$, define the entire zero-free symmetric factor $\mathcal{E}_{X,\kappa}(z) := E_X(\frac12+\kappa+\mathrm{i}z)E_X(\frac12+\kappa-\mathrm{i}z)$ and $\Xi_{\mathcal{E}_{X,\kappa}}(z) := \Xi(z)\mathcal{E}_{X,\kappa}(z)$. For all $(d,t) \in \mathbb{R}^2$,*

$$|\mathcal{E}_{X,\kappa}(t-\mathrm{i}d)|^2 = \exp\left(-4\sum_{2\le n\le X}\frac{\Lambda(n)}{n^{1/2+\kappa}\ln n}\cosh(d\ln n)\cos(t\ln n)\right) > 0, \tag{5.1}$$

and the canonical transverse Wronskian of $\Xi_{\mathcal{E}_{X,\kappa}}$ decomposes as

$$W_{\mathcal{E}_{X,\kappa}}(d,t) = |\mathcal{E}_{X,\kappa}(t-\mathrm{i}d)|^2\left[W(d,t) - 2|\Xi(t-\mathrm{i}d)|^2\sum_{2\le n\le X}\frac{\Lambda(n)}{n^{1/2+\kappa}}\sinh(d\ln n)\cos(t\ln n)\right]. \tag{5.2}$$

In particular, $\Xi_{\mathcal{E}_{X,\kappa}}(z_0) = 0 \iff \Xi(z_0) = 0$, and at every zero $z_0 = t_0 - \mathrm{i}d_0$ of Ξ one has $W_{\mathcal{E}_{X,\kappa}}(d_0,t_0) = |\mathcal{E}_{X,\kappa}(z_0)|^2W(d_0,t_0) = 0$.

Proof. Since $\frac12+\kappa+\mathrm{i}(t-\mathrm{i}d) = \frac12+\kappa+d+\mathrm{i}t$ and $\frac12+\kappa-\mathrm{i}(t-\mathrm{i}d) = \frac12+\kappa-d-\mathrm{i}t$, we have $n^{-(1/2+\kappa+\mathrm{i}z)} + n^{-(1/2+\kappa-\mathrm{i}z)} = n^{-1/2-\kappa}[n^{-d}\mathrm{e}^{-\mathrm{i}t\ln n} + n^d\mathrm{e}^{\mathrm{i}t\ln n}]$ whose real part is $2n^{-1/2-\kappa}\cosh(d\ln n)\cos(t\ln n)$. Taking $|\exp(\cdot)|^2 = \exp(2\mathrm{Re}(\cdot))$ gives (5.1). Applying the product rule $\frac12\partial_d(|\Xi|^2|\mathcal{E}_{X,\kappa}|^2) = |\mathcal{E}_{X,\kappa}|^2[W + |\Xi|^2\cdot\frac12\partial_d\ln|\mathcal{E}_{X,\kappa}|^2]$ and differentiating (5.1) with respect to d yields (5.2). □

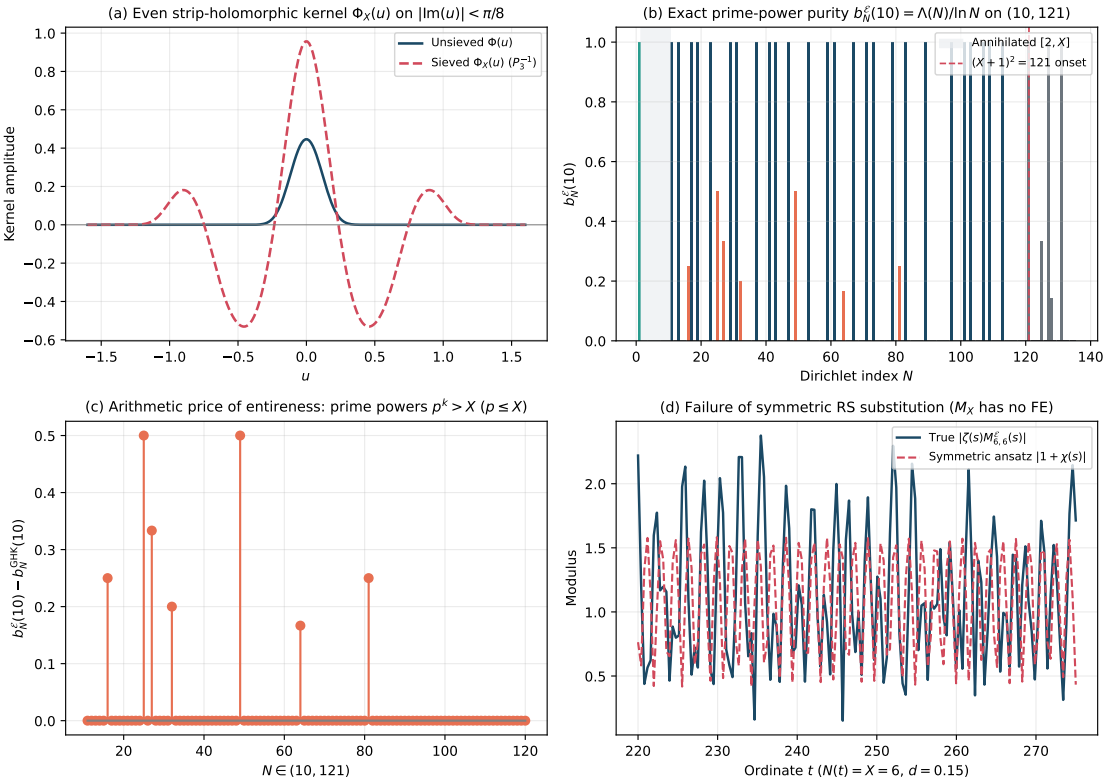


FIGURE 1. **(a)** Even strip-holomorphic sieved kernel $\Phi_X(u)$ vs. unsieved $\Phi(u)$. **(b)** Exact annihilation on $[2, 10]$ and prime-power purity $b_N^{\mathcal{E}}(10) = \Lambda(N)/\ln N$ on $(10, 121)$. **(c)** Fractional prime-power difference $b_N^{\mathcal{E}}(10) - b_N^{\mathrm{GHK}}(10)$ at $p^k > 10$ $(p \le 10)$. **(d)** Failure of naive symmetric Riemann–Siegel substitution $|1+\chi(s)|$ vs. true $|\zeta(s)M_{6,6}^{\mathcal{E}}(s)|$.

TABLE 2. 50-decimal benchmarks of the critical-strip obstructions (Propositions 4.1–4.3): asymmetric RS error $|\zeta M_{6,6}^{\mathcal{E}} - (1 + \chi)|$ at $t = 250$, stall of $X^{-2d}/(2d \ln X)$ at zero-detection depth $d = 1/\ln t$, and divergence of $w_2^*(d_0) = -(2^{2d_0} - 1)^{-1}$ as $d_0 \to 0^+$.

Proposition 4.2(ii): Stall at $d = 1/\ln t$ $(X = \sqrt{t/2\pi})$				Proposition 4.3: Single-Line Weight $w_2^*(d_0)$		
Height t	$d = 1/\ln t$	X^{-2d}	$X^{-2d}/(2d \ln X)$	Tuning d_0	$w_2^*(d_0)$	Off-line mismatch $(d = 2d_0)$
10^2	0.21715	0.54831	0.91247	0.600	−0.77077	2.29740
10^3	0.14476	0.48001	0.65402	0.250	−2.41421	1.41421
10^6	0.07238	0.42022	0.48470	0.050	−13.93273	1.07177
10^{12}	0.03619	0.39318	0.42120	0.001	−720.84764	1.00139

6. Scope, Boundary Conditions, and Open Questions

We summarize the exact mathematical scope established in this paper:

(1) *Unconditional identities on $\sigma > 1$ and in kernel space.* Lemma 2.1, Proposition 2.2, Theorem 3.1, and Proposition 5.1 hold unconditionally. In particular, the prime-power purity window $b_N^{\mathcal{E}}(X) = \Lambda(N)/\ln N$ on $(\lfloor X \rfloor, (\lfloor X \rfloor + 1)^2)$ and universal contractivity

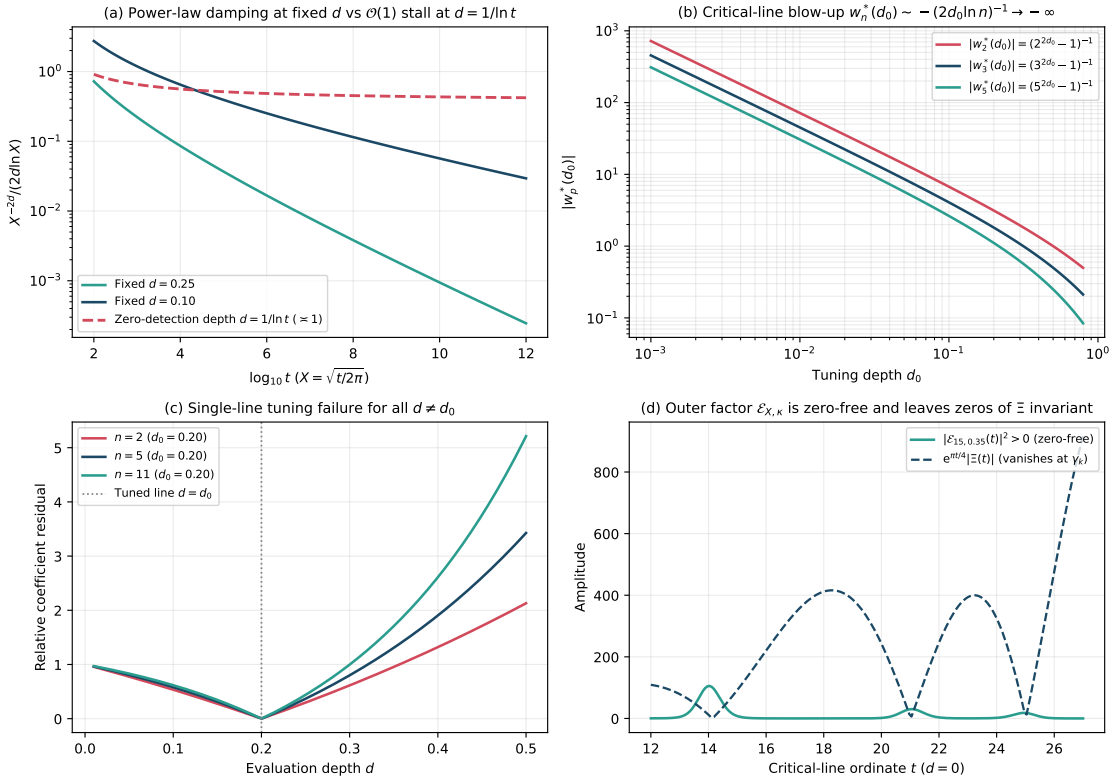


FIGURE 2. **(a)** Power-law decay of $X^{-2d}/(2d\ln X)$ at fixed d vs. $\mathcal{O}(1)$ stall along zero-detection depth $d = 1/\ln t$. **(b)** Critical-line divergence $|w_p^*(d_0)| \sim (2d_0 \ln p)^{-1} \to \infty$ as $d_0 \to 0^+$. **(c)** Off-line tuning failure of $w_n^*(d_0)$ for all $d \neq d_0$. **(d)** Strict positivity $|\mathcal{E}_{15,0.35}(t)|^2 > 0$ across critical zeros γ_k where $\Xi(\gamma_k) = 0$.

$0 \leq b_N^{\mathcal{E}}(X) \leq 1$ provide an exact arithmetic counterpart to Gonek–Hughes–Keating's X-rough indicator $b_N^{\mathrm{GHK}}(X)$ [2].

(2) *Why exponential prime gauges cannot eliminate critical-strip zeros.* By Proposition 4.1 and Proposition 5.1, multiplying $\Xi(z)$ by any entire zero-free factor $\mathcal{E}_{X,\kappa}(z)$ leaves the zero set of $\Xi(z)$ completely unchanged, and lack of a functional equation for $M_X(s)$ prevents symmetric sieving of both Riemann–Siegel branches simultaneously.

(3) *Mean-square mollification vs. pointwise control.* Controlling the mixed forward-and-reflected cross-terms of $|\zeta(\frac{1}{2}+it)M_X(\frac{1}{2}+it)|^2$ requires T-averaging via Selberg's mollifier method [10] ($X \leq T^{1/2-\varepsilon}$) or Levinson–Conrey shifted moments, rather than pointwise identities.

APPENDIX A. NUMERICAL AND INTERVAL CERTIFICATION PROTOCOLS

In the standalone verification suite `scripts/verify_paper6_sieved_mirror_prime_gauge.py`, all identities and obstruction benchmarks are verified at 50-decimal precision (`mpmath.mp.dps = 50`) and 35-digit directed-rounding interval arithmetic (`mpmath.iv.dps = 35`):

(1) *Exact recurrence and local generating function.* The Euler logarithmic-derivative recurrence $\ln N\, b_N^{\mathcal{E}}(X) = \sum_{d|N,d>X} \Lambda(d) b_{N/d}^{\mathcal{E}}(X)$ matches both the divisor sum $\sum_{k|N} c_k^{\mathcal{E}}(X)$ and the local generating product $\prod_{p^a\|N} \beta_a(k_p)$ to $< 10^{-48}$ across $1 \le N \le 150$.

(2) *Asymmetric Riemann–Siegel and interval blow-up checks.* At $(d,t) = (0.15, 250)$ ($N(t) = X = 6$), $|\zeta(s)M_{6,6}^{\mathcal{E}}(s) - (1+\chi(s))| = 0.595739 > 0.1$, certifying the failure of naive symmetric substitution, while interval arithmetic encloses $w_2^*(0.001) \in [-720.84764, -720.84763]$.

Use of AI and computational verification

In accordance with editorial policies on transparency in scientific computing, the author discloses that the AI research assistant **Gemini** (Google DeepMind) assisted in comparing the Gonek–Hughes–Keating Euler truncation against $E_X(s)$, formalizing the critical-strip functional-equation and Selberg explicit-formula obstructions, and implementing `scripts/verify_paper6_sieved_mirror_prime_gauge.py` (exit code 0).

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